

The Microcirculation and Lymphatic System

The Microcirculation

- Portion of the **systemic** and **pulmonary circulations** adapted for **exchange** of water, gases, nutrients, and metabolic wastes.
- Highly complex and least known.
- One cell thick and largest cross-sectional area.
- Lowest pulse pressure
- **Variable distribution:**
 - **high density** in **metabolically active tissues** such as heart, skeletal muscle, and glandular structures.
 - **low density** in **metabolically less active tissues** such as subcutaneous tissue, cornea or cartilage.

Microcirculation, cont'd

- Contains about 5% of the blood at any one time (most important part of blood).
- Most important part of CVS because it is the site where the ultimate purpose is realized.
- Closed system but with walls that are much more permeable than any other part of CVS.
- Because of great cross-sectional area, flow velocity is the least in the CVS.
- Slowing of the stream and dispersal over a maximum surface area makes the microcirculation particularly suitable for exchange.

Microcirculation, cont'd

Structural specialization for transport

Arterioles....metarterioles...capillaries...venules (fig..).

- The capillary is the principal site for the exchange of gases, water, nutrients, and waste products.
- In most instances capillary flow exclusively serves nutritional needs.
- In a few tissues, however, a large portion of the capillary flow is non-nutritional.
 - For example, in the glomeruli of the kidneys, the capillary flow forms the glomerular filtrate; capillaries also serve in temperature regulation of skin, and in delivery of hormones, and platelets.

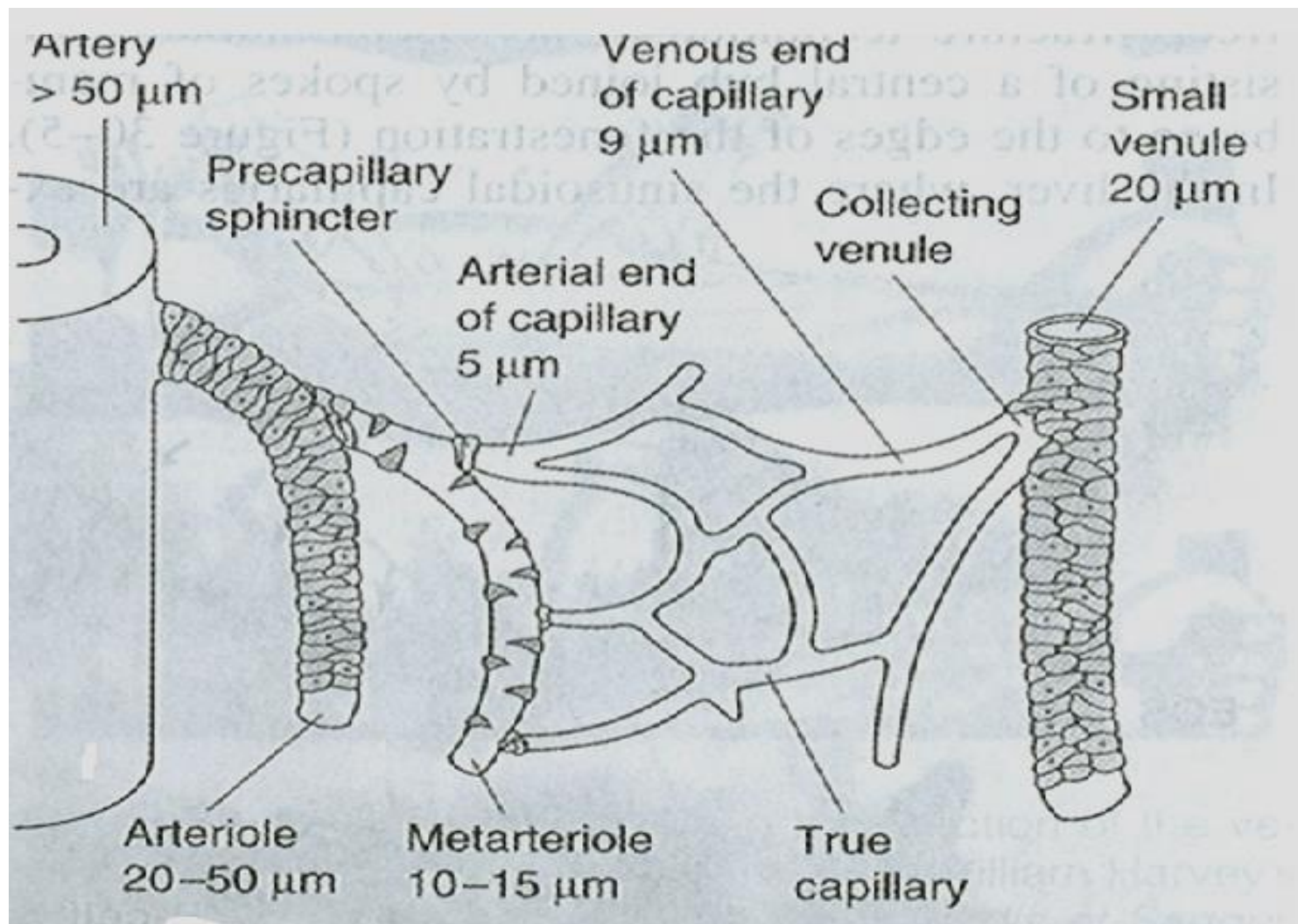


Fig..The Microcirculation. Arterioles give rise to metarterioles metarterioles to capillaries and capillaries to venules.

Capillaries

- ❑ Anatomically and functionally variable depending on special role of respective tissue or organ.
- ❑ Based on the continuity of their filtration barriers, capillaries have been labelled as continuous, fenestrated, and discontinuous.

1. Continuous capillaries:

- No recognizable intercellular openings.
- A subgroup with “low” or flattened cellular type is seen mainly in the muscle, nerve and adipose tissue.
- A second subgroup with a “high” or cuboidal epithelial structure seen in lymph nodes and thymus gland.

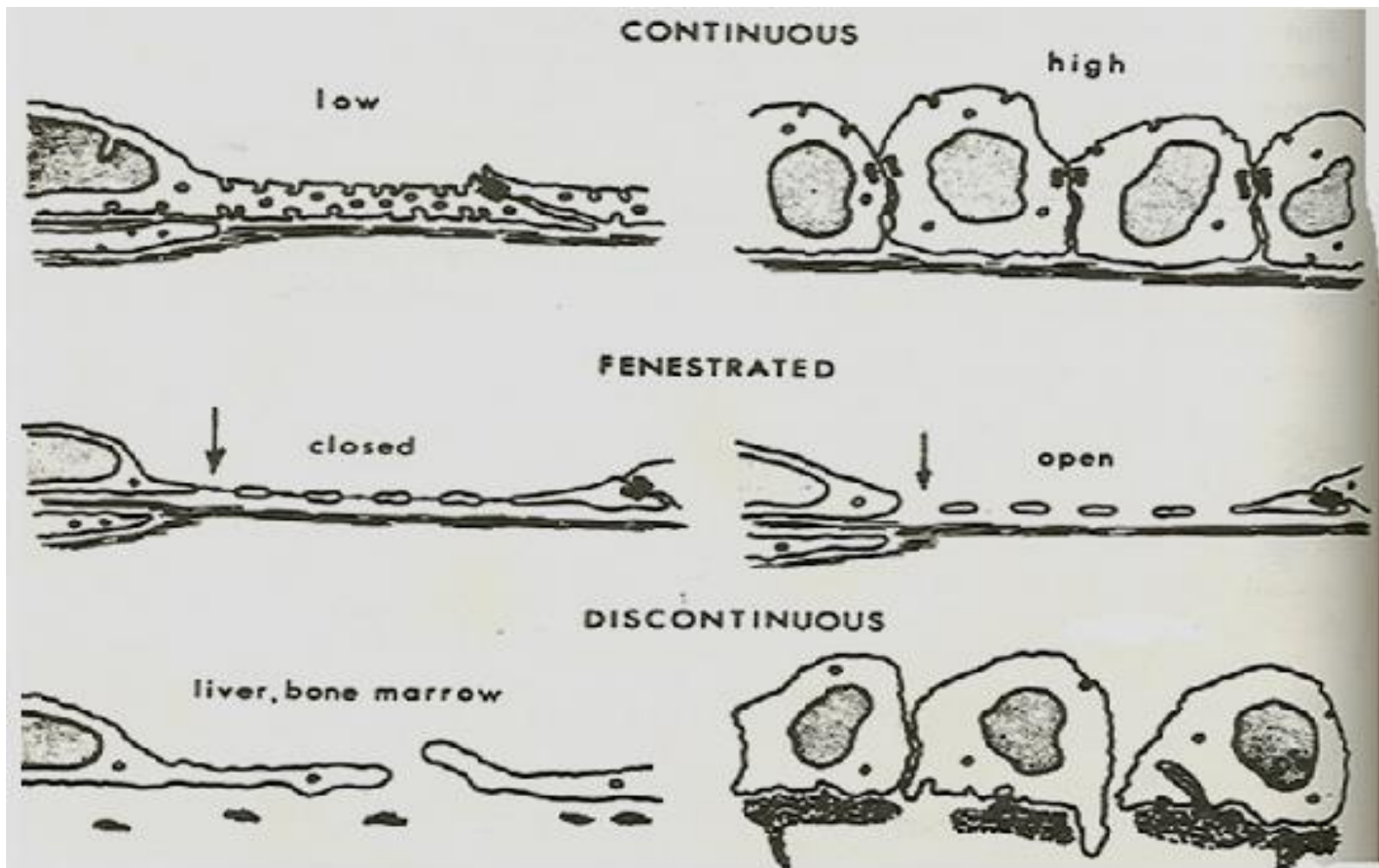


Fig..Continuous, fenestrated, and discontinuous types of capillary wall

2. Fenestrated capillaries

- Have small gaps which are either “closed” as in endocrine glands and intestinal villi or “open” as in renal glomeruli.
- Gaps range from 800-1000 Å; i.e. 15-20 times wider than continuous capillaries.

3. Discontinuous capillaries

- Endothelium with large intercellular gaps (**sinusoids**)-typical of the **liver, bone marrow and spleen**.
- Even the continuous capillaries with “tight junctions” are sufficiently wide to permit rapid diffusion of small molecular substances.
- The other two types have much greater permeability.
- Pressures in the capillaries vary considerably.
- Their total contribution to TPR is only 27%.

Trans-capillary Exchange

- Transport across capillaries is influenced by several factors:
 - i) type and molecular size of substance to be transported.
 - ii) balance of hydrostatic & colloid osmotic forces across the membrane.
 - iii) capillary surface available for exchange.
 - iv) physical characteristics of capillary membrane.
- Two major types of transport mechanisms that occur in capillary membrane include
 - diffusion and filtration/absorption.
 - Vesicular transport is a 3rd transport mechanism but is less common.

Cont'd

- 1) Diffusion:** most important trans-capillary exchange mechanism.
- **Lipid soluble materials** including O_2 , CO_2 and anaesthetic agents diffuse rapidly.
 - **Water-soluble substances** are limited to diffusion through intercellular slit-pores in which molecular movement is made possible by filtration.
 - Plays a much greater role in the exchange of **solutes**.

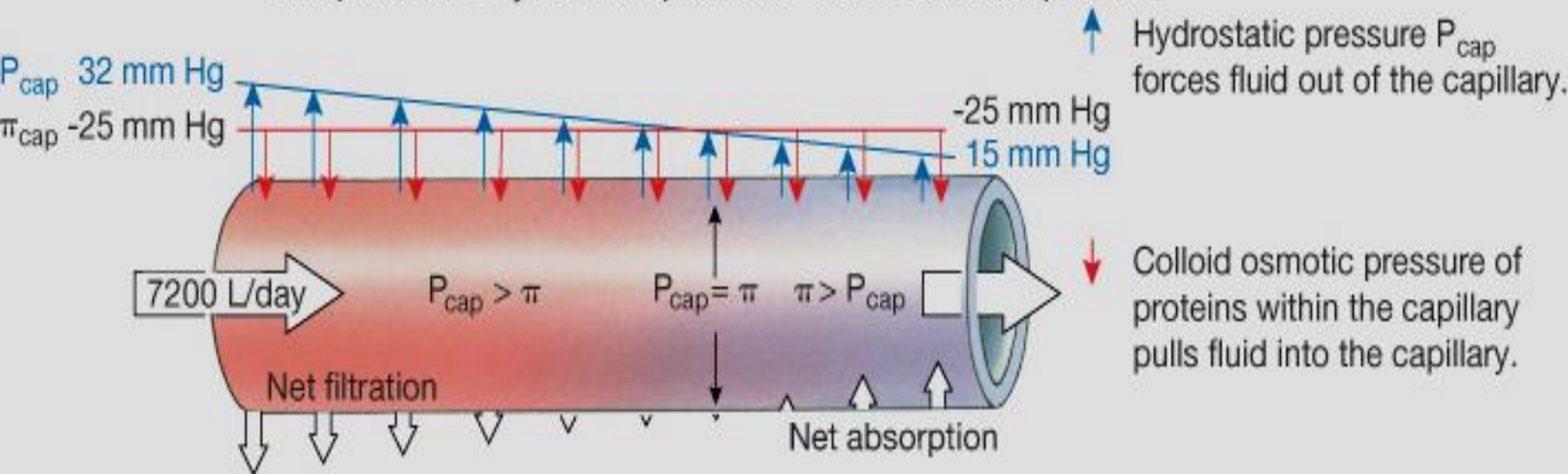
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2) Filtration-Absorption

- 20 litres of fluid filtered (excluding kidneys) into interstitium in 24 hrs.
- 18 L/day of this fluid is reabsorbed.
- The remaining 2 L/day make up the lymph flow that returns to the blood stream.
- Filtration-absorption of materials is dependent on the net d/c b/n hydrostatic pressures on each side of the membrane (Starling's law of the capillary).
- The driving force includes a hydrostatic pressure of about 32 mm Hg in arteriolar, about 15 mm Hg at the venular end and a plasma osmotic pressure (oncotic pressure mainly due to albumin) of 25 mm Hg.

Trans-capillary Exchange

(a) Filtration in systemic capillaries

$$\text{Net pressure} = \text{hydrostatic pressure} - \text{colloid osmotic pressure}$$


Net flow out = 2 L/day Lymph flow

NB: Filtered=20L/day
Reabsorbed =18L/day

KEY

↓ π = Colloid osmotic pressure

↑ P_{cap} = Capillary hydrostatic pressure

- At the arteriolar end, the pressure difference (32-25 mm Hg) is **positive** and **facilitates filtration**; at venular end, the pressure difference is **negative**, and hence **absorption** occurs here (fig..).
- The mathematical approximation for filtration = Hydraulic drive – osmotic suction or **$F = K(S)(P_c - P_i) - (\pi_c - \pi_i)$** ,
 where F = net movement of fluid, K= specific permeability constant, S = capillary surface area, P_c = capillary pressure, P_i = interstitial fluid pressure, π_c = capillary colloid osmotic pressure, π_i = interstitial colloid osmotic pressure.

At Arterial Side of Capillary

Filtration forces (mmHg)	Reabsorption forces (mmHg)
Capillary hydrostatic pressure = 35	Plasma colloid osmotic pressure = 25
Tissue fluid colloid osmotic pressure = 3	Interstitial hydrostatic pressure = 1
Total = 38	Total = 26
Balance of forces = $38 - 26 = 12$ mmHg	

At Venous Side of Capillary

Filtration forces (mmHg)	Reabsorption forces (mmHg)
Capillary hydrostatic pressure = 15	Plasma colloidal osmotic pressure = 25
Tissue fluid colloidal osmotic pressure = 3	Interstitial hydrostatic pressure = 1
Total = 18	Total = 26
Balance of forces = $26 - 18 = 8$ mmHg	

3) Vesicular Transport

- Involves an element of **active transport**.
- **Very slow** and contributes **very little** to total capillary exchange.
- Cell mass forms in-pockets that surround substances in the vicinity of the wall.
- As shown in fig., in-pocketing **on cell surface** → vesicles that completely engulf substances.
- Vesicles then break off and migrate through the cytoplasm to the opposite wall where the process reverses (out-pocketing) itself and the contents of the vesicle are released to the outside.
- Accounts for transport of **large molecular substances**.

c) Transport of macromolecules

1. Endocytosis:

- Engulfing of materials by invaginating the outer part of a cell membrane until it buds off within the cytoplasm
- Requires energy

1.1. Phagocytosis: –“cell eating .”

- **Very** large molecules such as bacteria, dead RBC, etc surrounded by cell membrane and taken up

1.2. Pinocytosis - cell invagination occurs into cell and pinches off to form boundary of an intracellular vesicle, vacuole or tubule. e.g. absorption of **undigested protein in gut of newborns**.

- Requires energy

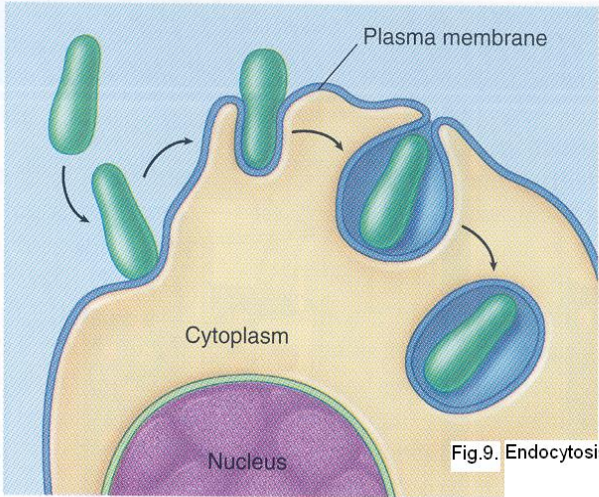
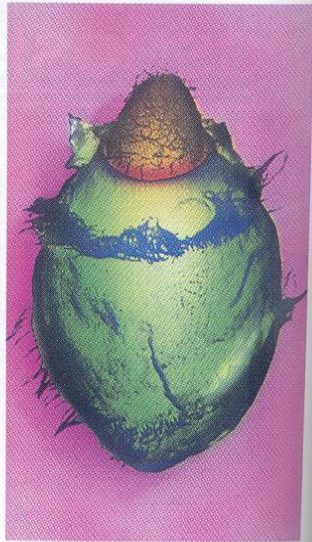


Fig.9. Endocytosis

(a) Engulfing of relatively large organism or fragment of organic matter



(b)

protist didinium nasutum ingesting protist paramecium

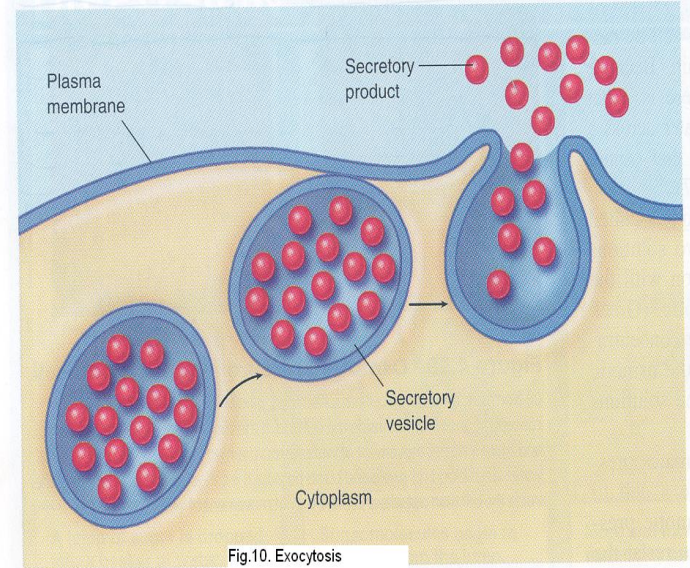


Fig.10. Exocytosis

Regulation of the Microcirculation

1. Neural

2. Metabolic

- Flow through arterioles & met-arterioles is mainly controlled by **sympathetic vasoconstriction**.
- Flow through capillaries is controlled by the **pre-capillary sphincters**.
- Pre-capillary sphincters are influenced by **local metabolic state** of tissue – **auto-regulation**.

Regulation of Blood Flow

- Altered demands of flow are met either by a change in central perfusion pressure or by adjustment of vessel calibre.
- Vessel calibre is adjusted by two general mechanisms:
 1. Local regulation
 2. Central regulation (fig..below)

1. Local (intrinsic) Systems of Regulation

a) Myogenic Hypothesis

- $\uparrow \text{BF} \rightarrow \uparrow$ stretch of blood vessels $\rightarrow$ constriction to normalize blood flow.
- $\downarrow \text{BF} \rightarrow \downarrow$ tension of blood vessel $\rightarrow$ relaxation to normalize BF.

b) Metabolic Hypothesis

$\downarrow$ MR or $\downarrow$ O₂ → production of vasodilator or vasoconstrictor metabolites

Examples:

- 1) $\downarrow$ O₂, $\downarrow$ H⁺; cause relaxation or constriction of blood vessels.
- 2) Adenosine...dilatation of arterioles.
- 3) $\uparrow$ CO₂ → moderate vasodilatation in most tissues but marked vasodilatation in brain.
 $\uparrow$ CO₂ via sympathetic supply → $\uparrow$ VMC activity → powerful vasoconstriction.

- 4) K^+ → Inhibition of smooth muscle contraction → vasodilatation
- 5) Mg^{++} → same effect as K^+
- 6) Na^+ → ↑ Osmolality of body fluids → arteriolar dilatation
- 7) Lactate → Vasodilatation
- 8) Ca^{++} → Smooth muscle contraction → vasoconstriction, etc.

c) Tissue Pressure Hypothesis

↑Perfusion pressure → ↑blood volume → net transfer of fluid from intravascular to extra-vascular space → ↑tissue pressure (turgor) → compression on thin-walled blood vessels

Extrinsic (Remote) Regulation

a) Neural Regulation

ANS receiving impulses from higher centres combines transmitters and receptors to change blood flow through different parts of the body as shown in fig. below:

b) Hormonal Regulation

1. Noradrenaline
2. Adrenaline
3. Angiotensin
4. Kinins: bradykinin
5. Prostaglandins
6. ADH

c) Balance b/n intrinsic & extrinsic regulation of BF

This is important for directing BF to areas of greater need.

1) Brain and Heart

- Tolerance for decreased BF is limited
- Intrinsic flow-regulating mechanism is dominant

2) Skin

- Neural control is dominant
- Has some local control

3) Skeletal muscle

- Neural control is dominant
- Start or anticipation of exercise → activation of cholinergic sympathetic dilator system
- After onset of exercise → intrinsic regulation

Introduction

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- **The lymphatic system** consists of a **fluid** called **lymph**, **vessels** called **lymphatic vessels**.
- A number of structures & organs containing **lymphatic tissue** (lymphocytes within a filtering tissue), and **red bone marrow**.
- The LS assists in **circulating body fluids** & helps **defend** the body against disease-causing agents.
- Most components of **blood plasma filter** through blood capillary walls to form **interstitial fluid**.
- After interstitial fluid passes into lymphatic vessels, it is called **lymph**.
 - **Interstitial fluid** is found between cells, and **lymph** is located within the lymphatic vessels and lymphatic tissue.

- The LS represents an **accessory route** through which fluid can flow from interstitial spaces into the blood.
- Most important, the lymphatics can **carry proteins** and **large particulate matter** away from the tissue spaces, neither of which can be removed by absorption directly into the blood capillaries.
- This return of proteins to the blood from interstitial spaces is an essential function **without which we would die within about 24 hours.**

Lymph Channels of the Body

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- Almost all tissues of the body have special **lymph channels** that **drain excess fluid** directly from interstitial spaces.
- The exceptions include **superficial portions of skin, central nervous system, endomysium** of the **muscles**, and **bones**.
- But, even these tissues have **minute interstitial channels** called **pre-lymphatics** through which interstitial fluid can flow; this fluid eventually empties either into the **lymphatic vessels** or, in the case of brain, into the **cerebrospinal fluid** and then directly back into the blood.

- All the lymph vessels from the **lower part of the body** empty into the **thoracic duct**, which in turn empties into the venous system at the juncture of **left internal jugular vein** and **left subclavian vein**.
- Lymph from left side of the head, left arm, and parts of the chest region also enters the **thoracic duct** before it empties into the veins.
- Lymph from right side of the neck & head, right arm, and parts of the right thorax enters the **right lymph duct** (*much smaller than the thoracic duct*), which empties into the blood venous system at the juncture of **right subclavian vein** and **internal jugular vein**.

Lymphatics: Structure and Roles

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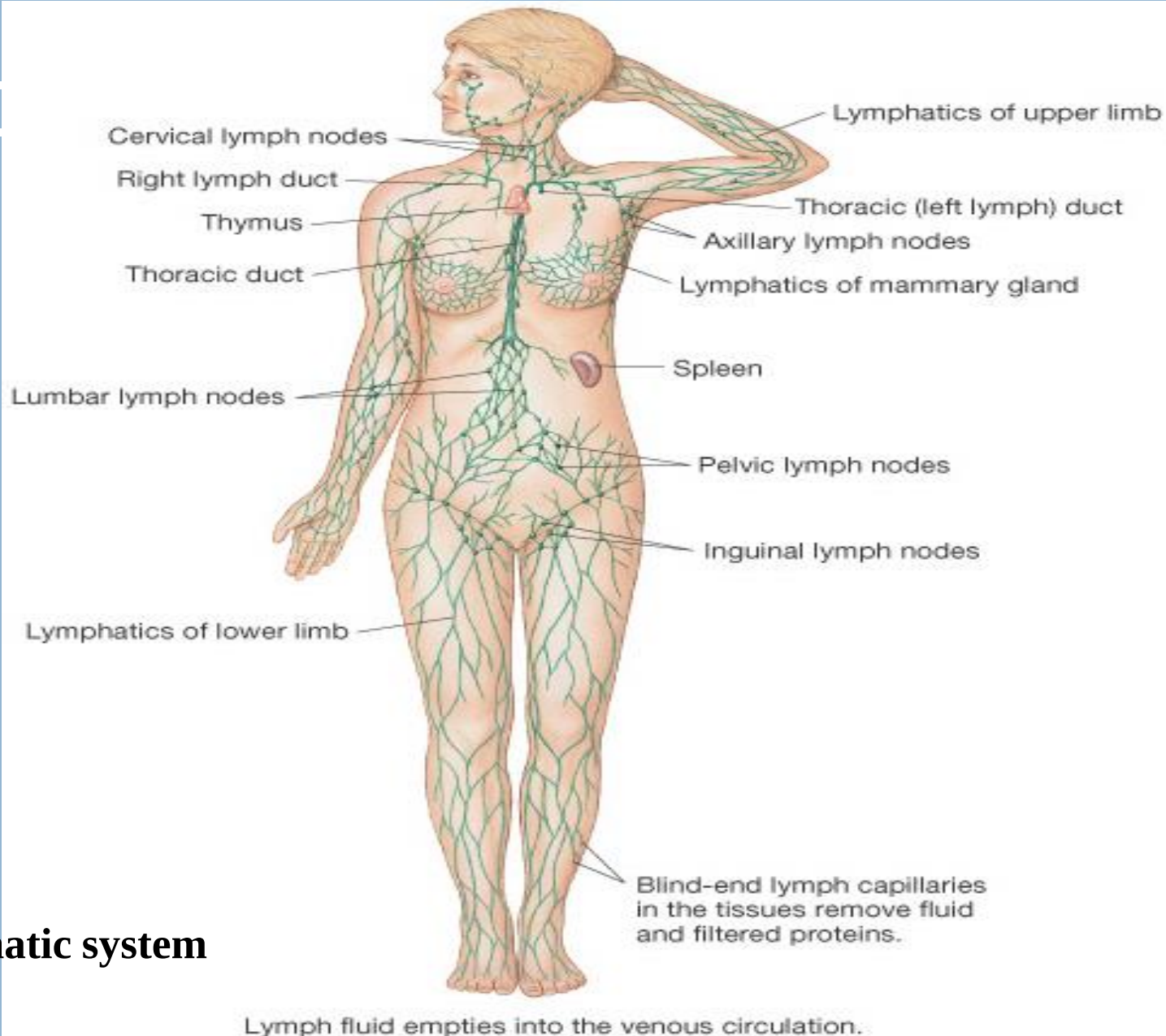
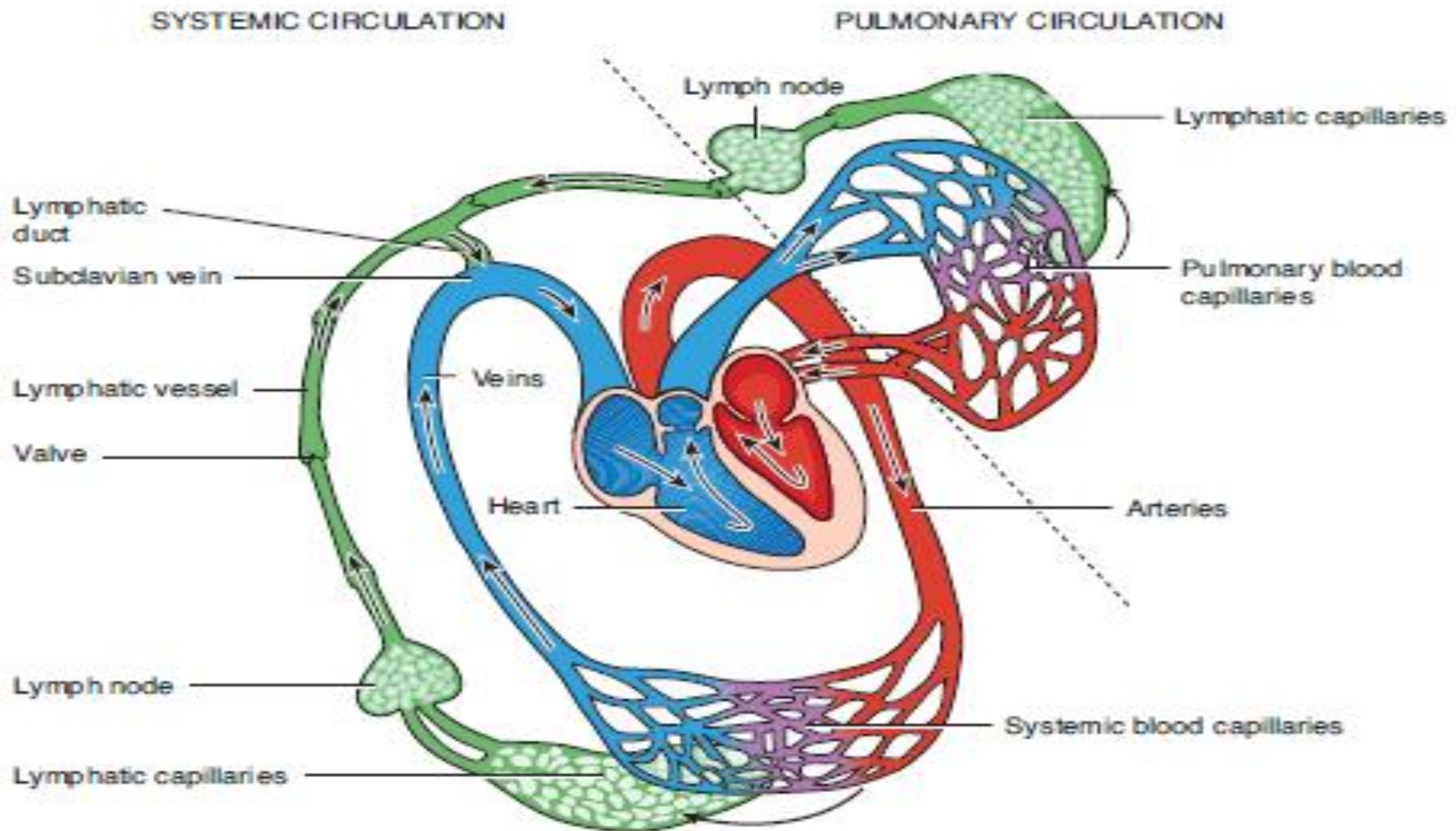


Figure: The lymphatic system

Schematic diagram showing the relationship of the lymphatic system to the cardiovascular system. Arrows indicate the direction of flow of lymph and blood



Terminal Lymphatic Capillaries and Their Permeability

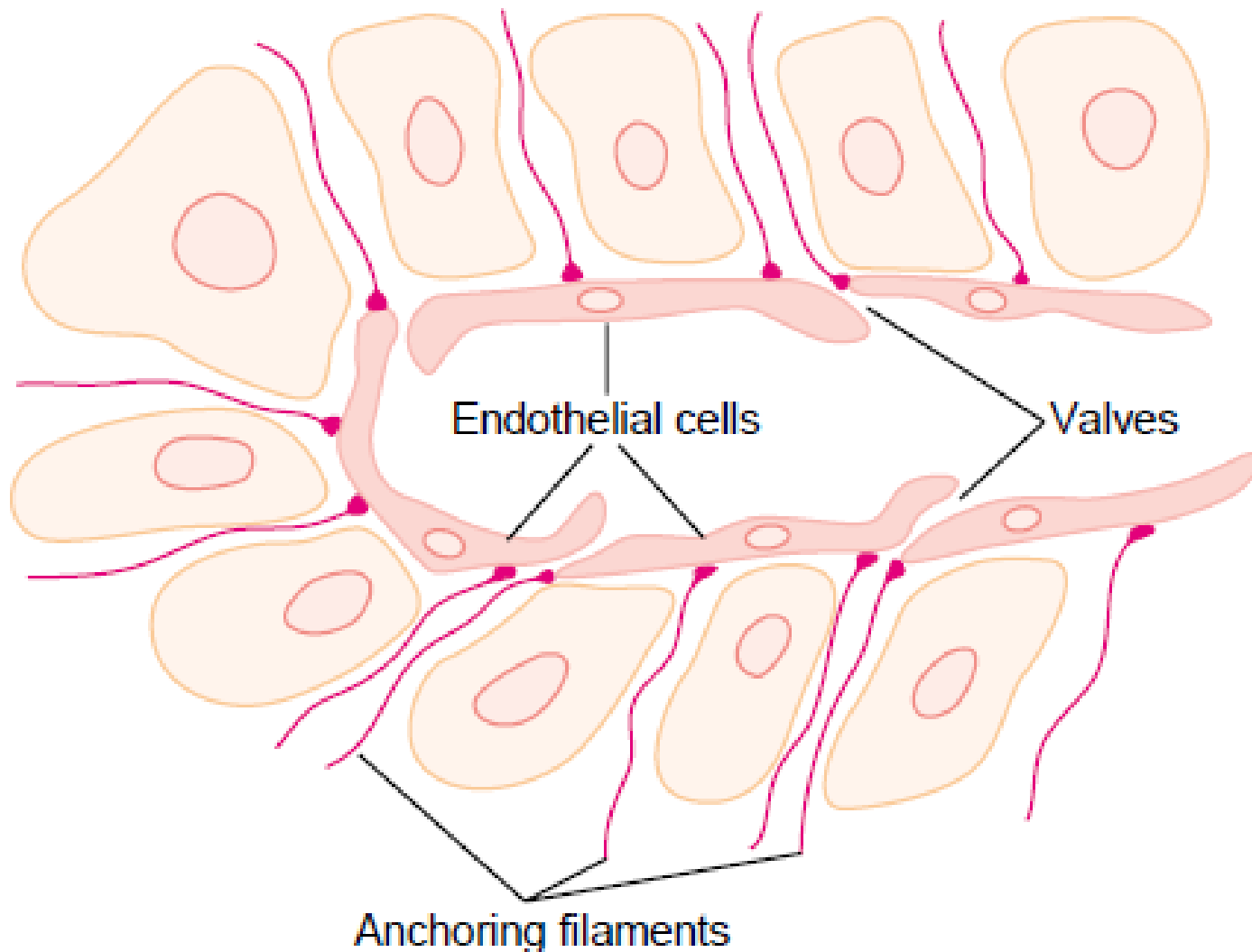
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- Most of the fluid filtering from the *arterial ends of the blood capillaries flows among the cells and finally is reabsorbed back into the venous ends of the blood capillaries*; but on average, about 1/10 of the fluid instead enters the *lymphatic capillaries and returns to the blood through the lymphatic system rather than through the venous capillaries*.
- The total quantity of all this lymph is normally only 2-3 L/day.

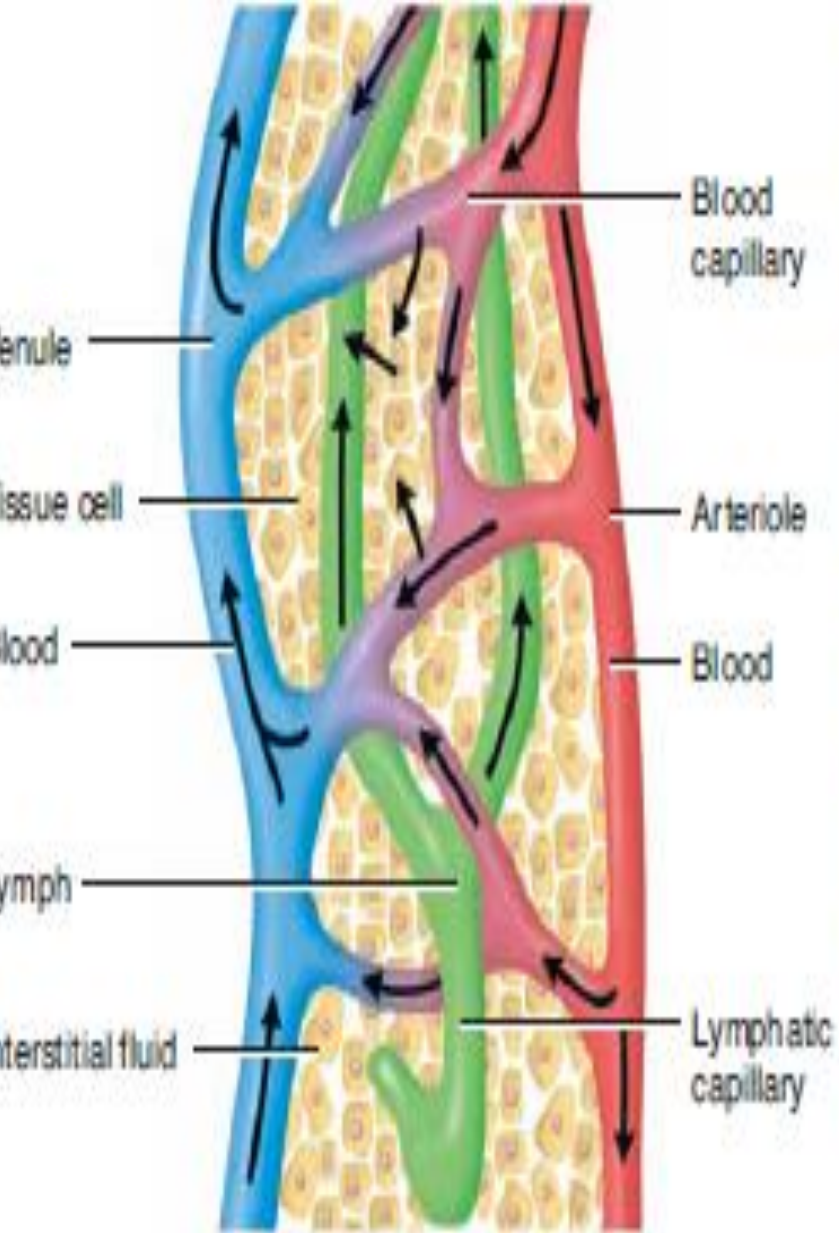
- The fluid that returns to the circulation by way of the lymphatics is extremely important because substances of high molecular weight, such as proteins, cannot be absorbed from the tissues in any other way, although they can enter the lymphatic capillaries almost unimpeded.
- At the junctions of adjacent endothelial cells, the edge of one endothelial cell overlaps the edge of the adjacent cell in such a way that the overlapping edge is free to flap inward, thus forming a **minute valve** that opens to the interior of the lymphatic capillary.

Special structure of the lymphatic capillaries that permits passage of substances of high molecular weight into the lymph.

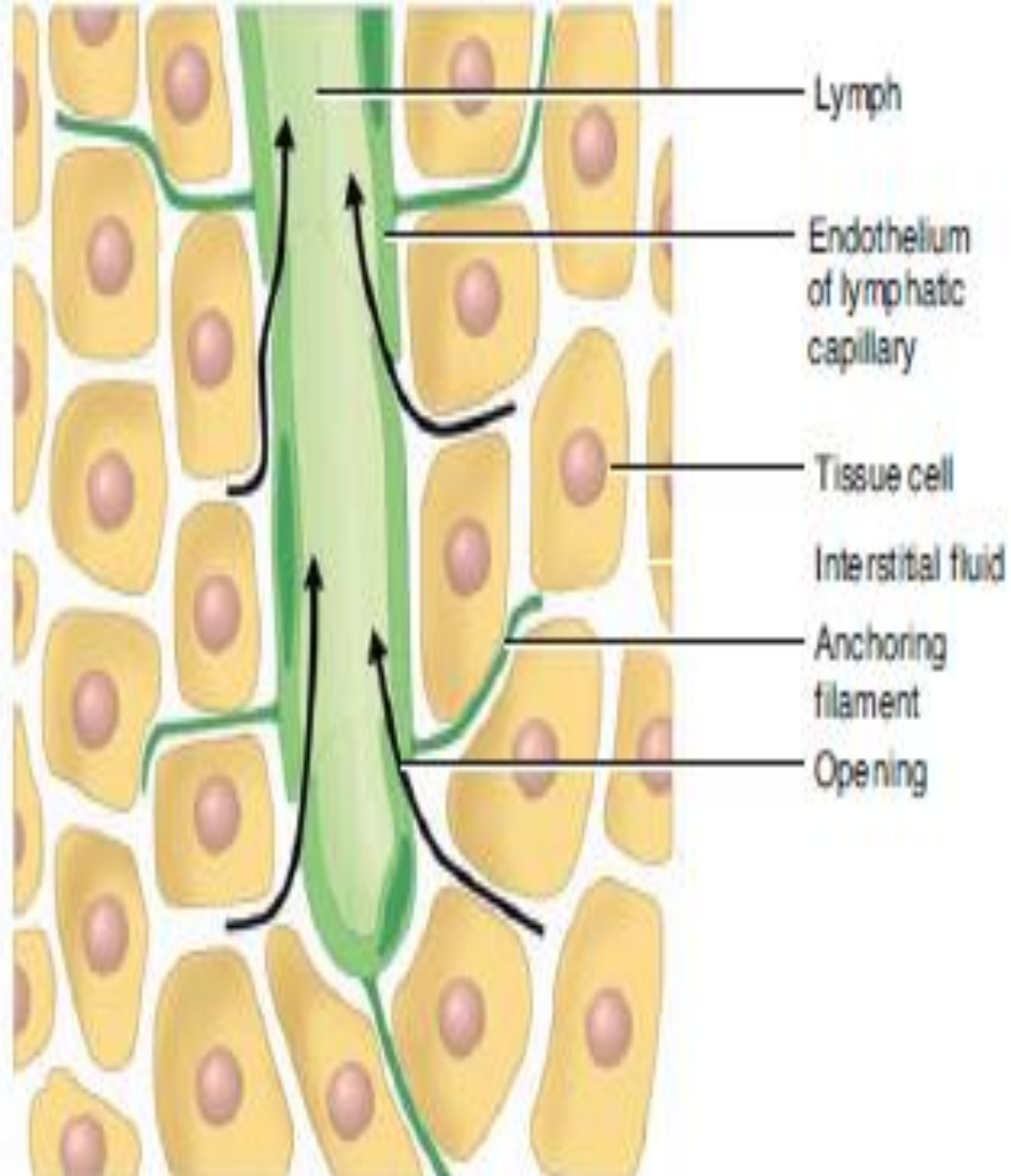
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- ❑ Interstitial fluid, along with its suspended particles, can push the valve open & flow directly into the lymphatic capillary.
- ❑ But this fluid has difficulty leaving the capillary once it has entered b/c any backflow closes the flap valve.
- ❑ The lymphatics have valves at the very tips of the terminal lymphatic capillaries as well as valves along their larger vessels up to the point where they empty into the blood circulation.



(a) Relationship of lymphatic capillaries to tissue cells and blood capillaries



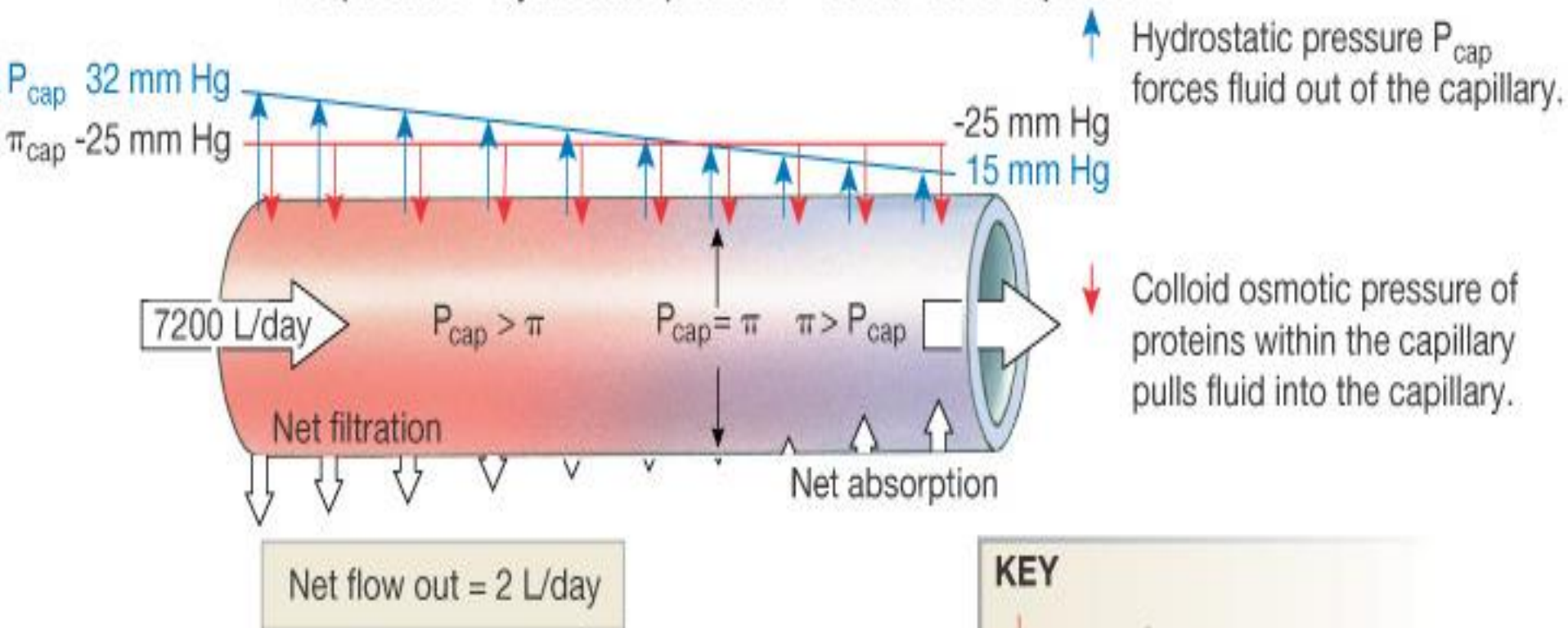
(b) Details of lymphatic capillary

Transcapillary Exchange

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(a) Filtration in systemic capillaries

Net pressure = hydrostatic pressure – colloid osmotic pressure



KEY

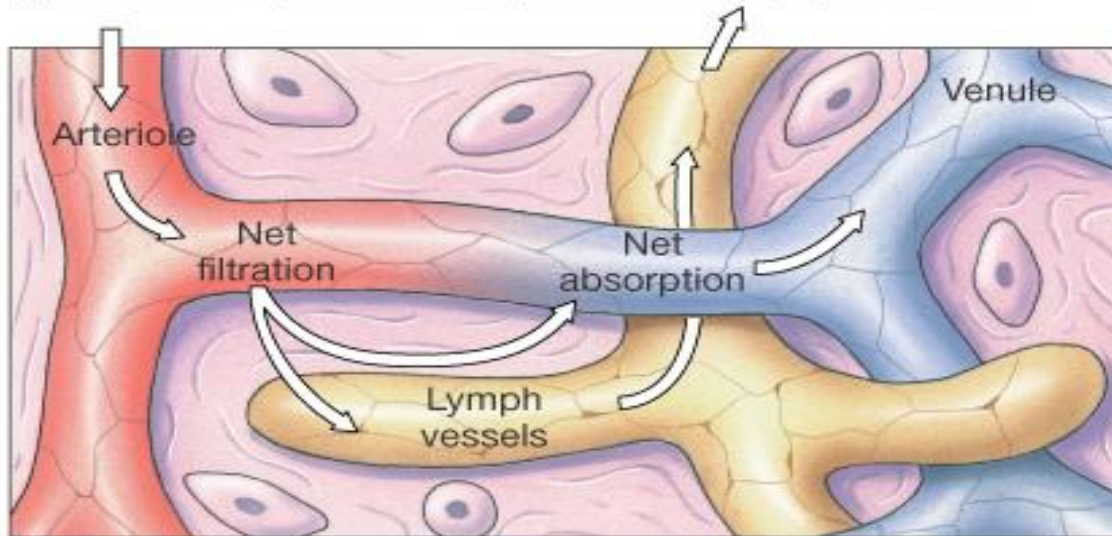
- ↓ π = Colloid osmotic pressure
- ↑ P_{cap} = Capillary hydrostatic pressure

Net outflow into ECF

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- Net filtration – net absorption = net outflow
- About 2 L/day collected by lymph vessels

(b) Relationship between capillaries and lymph vessels

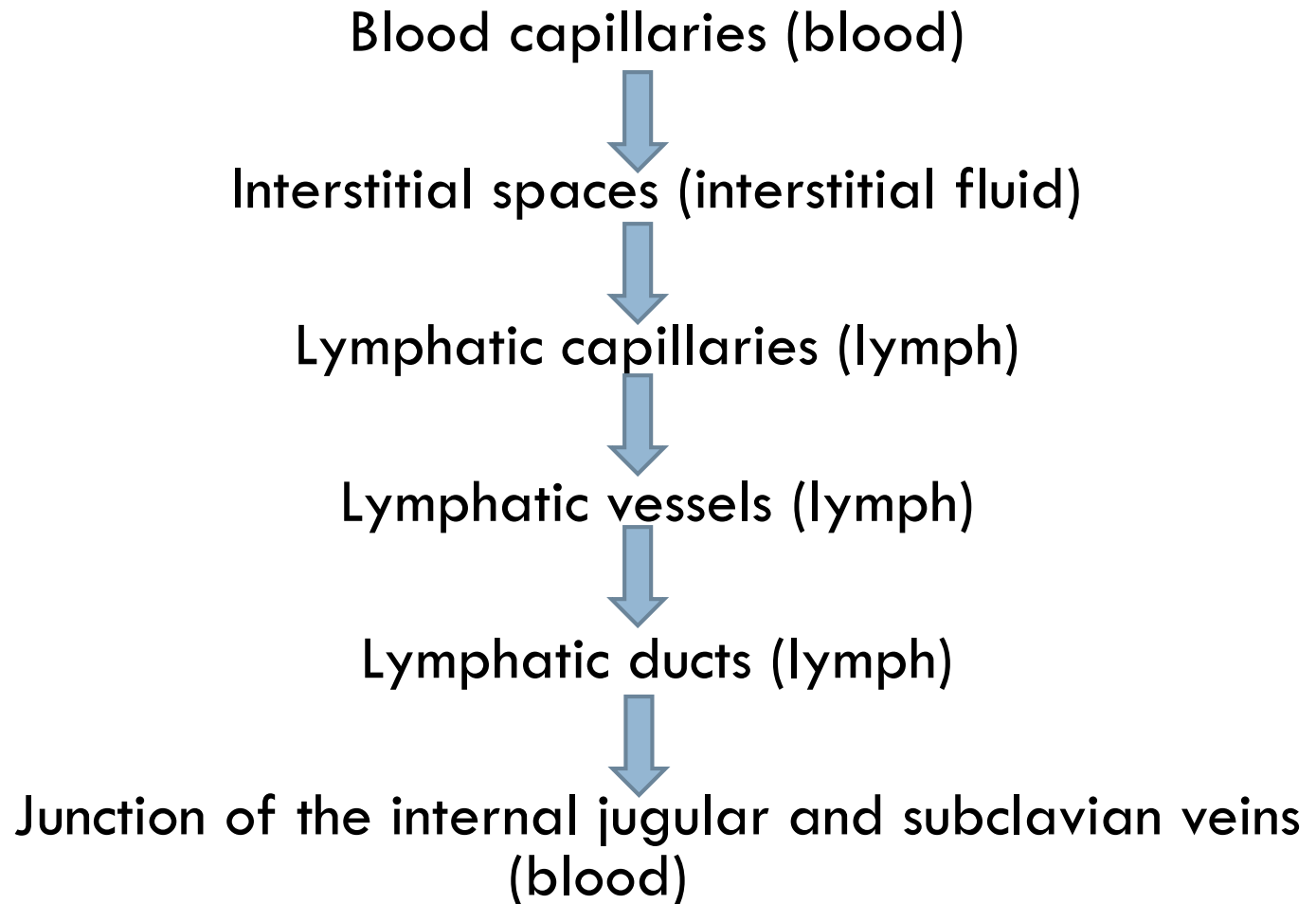


The excess water and solutes that filter out of the capillary are picked up by the lymph vessels and returned to the circulation.

Figure: Fluid exchange at the capillary

The sequence of fluid flow

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Formation of Lymph

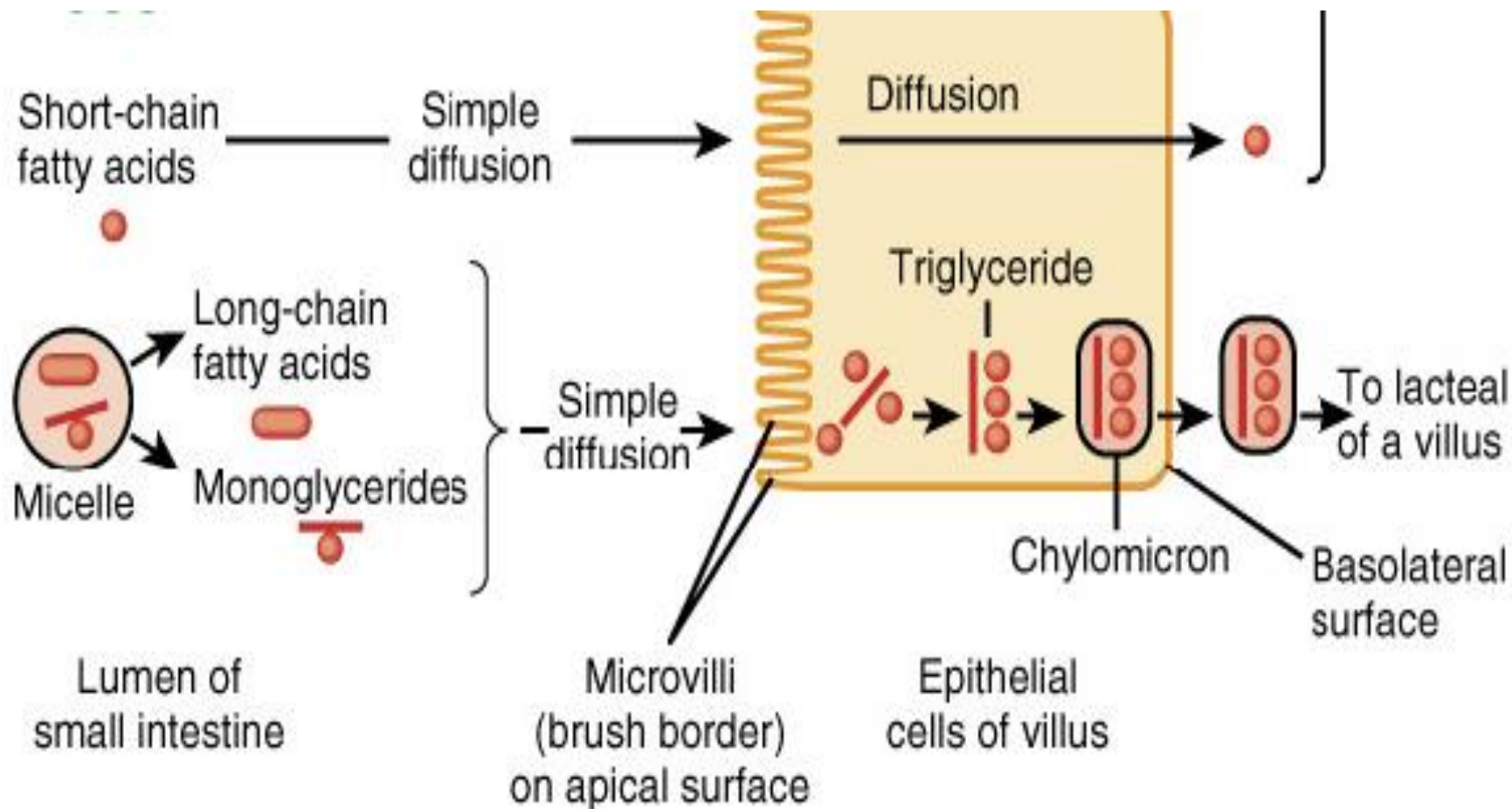
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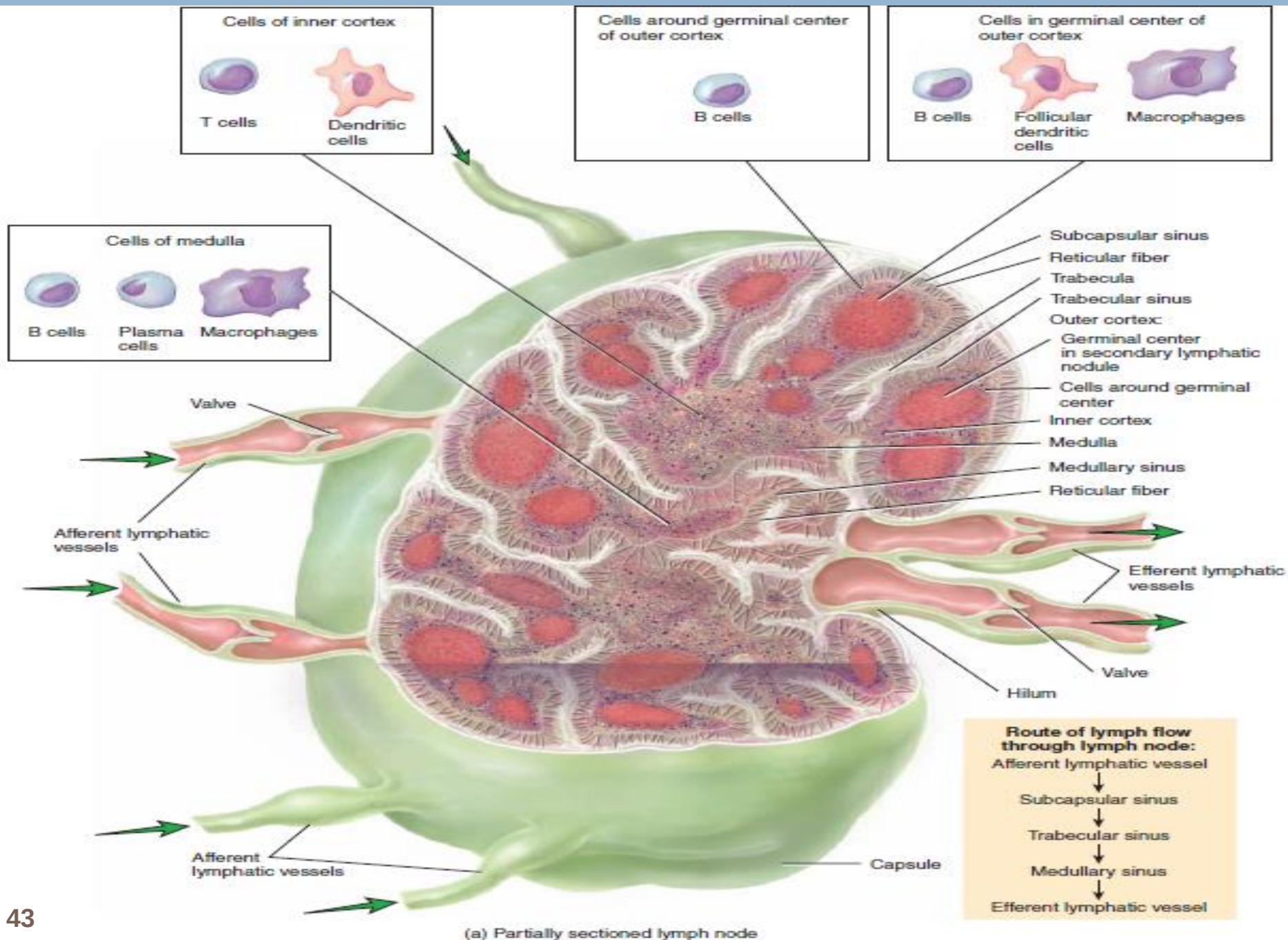
- Lymph is derived from interstitial fluid that flows into lymphatics.
- Lymph as it first enters the terminal lymphatics has almost the same composition as the interstitial fluid.
- The protein concentration in the interstitial fluid of most tissues averages about **2 g/dl**, and the protein concentration of lymph flowing from these tissues is near this value.
- Conversely, lymph formed in the liver has a protein concentration as high as **6 g/dl**, and lymph formed in the intestines has a protein concentration as high as **3 to 4 g/dl**.
- B/c about 2/3 of all lymph normally is derived from liver and intestines, **thoracic duct lymph**, which is a mixture of lymph from all areas of the body, usually has a protein conc. of **3 to 5 g/dl**.

Functions of the lymphatic system

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- **The lymphatic system** has **three primary functions**:
 1. ***Drains excess interstitial fluid.*** *Lymphatic vessels drain excess interstitial fluid from tissue spaces & return it to the blood.*
 2. ***Transports dietary lipids.*** *Lymphatic vessels transport lipids and lipid-soluble vitamins (A, D, E, and K) absorbed by GI tract.*
 3. ***Carries out immune responses.*** *Lymphatic tissue initiates highly specific responses directed against particular microbes or abnormal cells.*





- **Digested fats** are absorbed & then transported from the villi in the small intestine to the bloodstream via the lacteals and lymph vessels.
- **New lymphocytes** are manufactured in the lymph nodes;
- **Antibodies** (manufactured in the lymph nodes) assist the body to build up an effective immunity to infectious diseases;
- Lymph nodes play an important role in **defence mechanism** of the body.
 - ▣ They **filter out micro-organisms** (such as bacteria) and foreign substances such as toxins, etc.
- **Transports large molecular compounds** (such as enzymes and hormones) from their manufactured sites to the bloodstream.

- The lymphatic system also plays a central role in controlling
 - ▣ Concentration of proteins in interstitial fluids.
 - It plays an important role in returning plasma proteins to the bloodstream.
 - ▣ Volume of interstitial fluid, and
 - ▣ Interstitial fluid pressure.

Rate of Lymph Flow

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- About **100 ml/hr** of lymph flows through the *thoracic duct* of a resting human, and approximately another **20 ml/hr flows** into the circulation through other channels, making a total estimated lymph flow of about **120 ml/hr** or **2 to 3 L/day**.
- Normal lymph flow is very little at interstitial fluid pressure.
- Any factor that **increases interstitial fluid pressure** also increases lymph flow if the lymph vessels are functioning normally.
 - Elevated capillary pressure
 - Decreased plasma colloid osmotic pressure
 - Increased interstitial fluid colloid osmotic pressure
 - Increased permeability of the capillaries

Lymphatic Pump Increases Lymph Flow

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- When a collecting lymphatic or larger lymph vessel becomes stretched with fluid, the smooth muscle in the wall of the vessel automatically contracts.
- Furthermore, each segment of the lymph vessel b/n successive valves functions as a separate automatic pump.
- Even slight filling of a segment causes it to contract, and the fluid is pumped through the next valve into the next lymphatic segment.
- This fills the subsequent segment, and a few seconds later it contracts, the process continuing all along the lymph vessel until the fluid is finally emptied into the blood circulation.
- In a **very large lymph vessel such as the thoracic duct**, this lymphatic pump can generate pressures as great as 50 to 100 mm Hg.

Lymphatic pump caused by external intermittent compression of the lymphatics

- Contraction of surrounding **skeletal muscles**.
- **Movement of the parts** of the body.
- **Pulsations of arteries** adjacent to the lymphatics.
- **Compression of the tissues by objects** outside the body.
 - ❖ Lymphatic pump becomes very active during **exercise**, often increasing lymph flow 10-30 fold.
 - ❖ **During periods of rest**, lymph flow is sluggish, almost zero.

Lymphatic capillary pump

- When fluid enters the tissue & causes the tissue to swell, the anchoring filaments pull on the wall of the lymphatic capillary, and fluid flows into terminal lymphatic capillary through the junctions between the endothelial cells.
- Then, when the tissue is compressed, pressure inside the capillary increases and causes the overlapping edges of the endothelial cells to close like valves.
- Therefore, the pressure pushes the lymph forward into the collecting lymphatic instead of backward through the cell junctions.